
Telecom Subscriber Attrition Pattern Discovery

Sprint-1(EDA)

Industry: Telecommunications

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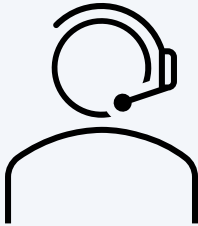
Organization: Anudip Foundation



Industry Overview

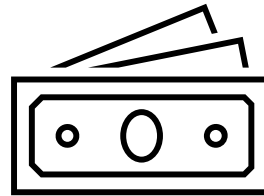
Selected Industry

Telecommunications: mobile, phone, internet and streaming-related services sold to consumers on contracts and subscriptions.



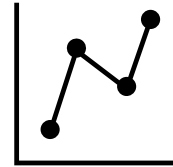
Introduction

A subscription-driven, highly competitive industry with large customer bases and recurring monthly revenue, where retaining customers is critical.



Why Data Analysis Matters

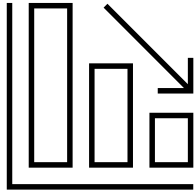
Customer, service, contract and billing data show which segments are associated with churn, helping plan retention.



Problem Statement

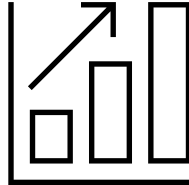
The Problem

Telecom companies lose revenue when customers discontinue their services. It is unclear which customer segments are most associated with churn.



Why It Matters

Churn reduces recurring revenue. Identifying at-risk segments supports targeted customer-retention planning.



Who Is Affected

The telecom company (management, marketing and retention teams) and customers who would benefit from better-targeted service.



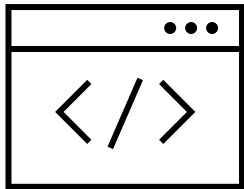
Project Objectives

- Examine customer demographics, service usage, contract, payment, tenure and billing data
- Measure the overall churn count and proportion
- Identify patterns associated with churn across customer segments
- Compare tenure and charges between churned and retained customers
- Support retention planning with descriptive, data-based insights

Proposed Solution & Analysis Questions

Proposed Solution

Use Python (Pandas, NumPy, Matplotlib, Seaborn) in Jupyter to clean the data, explore churn patterns by segment and visualize results. This is descriptive analysis: relationships shown do not prove causation and it is not a predictive model.



Analysis Questions

- What share of customers churned?
- How does churn differ by contract type?
- How does tenure relate to churn?
- Do monthly and total charges differ by churn status?
- How does churn vary by payment method, internet service and demographics?
- How do service add-ons relate to churn?

Dataset Information

Dataset Name	Telco Customer Churn (raw_dataset.csv)
Source	Hugging Face: scikit-learn/churn-prediction
Rows	7,043 (after cleaning)
Columns	21
File Format	CSV
Important Features	tenure, Contract, PaymentMethod, MonthlyCharges, TotalCharges, InternetService, SeniorCitizen, Partner, Dependents, tech support, online security, streaming services, Churn

Each row represents one telecom customer with demographic, service, contract, billing and churn-status information.

Data Collection & Understanding

- Collected: downloaded the public Telco churn dataset as a CSV
- Source: Hugging Face (scikit-learn/churn-prediction)
- Shape: 7,043 rows × 21 columns
- Data types: numeric and categorical; TotalCharges needed conversion to numeric
- Missing / duplicates: notebook checks found no duplicate or missing rows after cleaning
- Checks: shape, info, describe, missing and duplicate counts

Dataset Information:

```
<class 'pandas.DataFrame'>
```

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	customerID	7043 non-null	str
1	gender	7043 non-null	str
2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	str
4	Dependents	7043 non-null	str
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	str
7	MultipleLines	7043 non-null	str
8	InternetService	7043 non-null	str
9	OnlineSecurity	7043 non-null	str
10	OnlineBackup	7043 non-null	str
11	DeviceProtection	7043 non-null	str
12	TechSupport	7043 non-null	str
13	StreamingTV	7043 non-null	str
14	StreamingMovies	7043 non-null	str
15	Contract	7043 non-null	str
16	PaperlessBilling	7043 non-null	str
17	PaymentMethod	7043 non-null	str
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	float64
20	Churn	7043 non-null	str

dtypes: float64(2), int64(2), str(17)

memory usage: 1.1 MB

Data Cleaning & Transformation

- Trimmed whitespace in text fields
- Converted TotalCharges to numeric values
- Filled blank charge values with zero
- Converted tenure to numeric
- Removed exact duplicate rows
- Saved output as Dataset/cleaned_dataset.csv
- Script: Python/data_cleaning.py

```
1 # 4. Check duplicate records
2 print("Duplicate rows:", df.duplicated().sum())
```

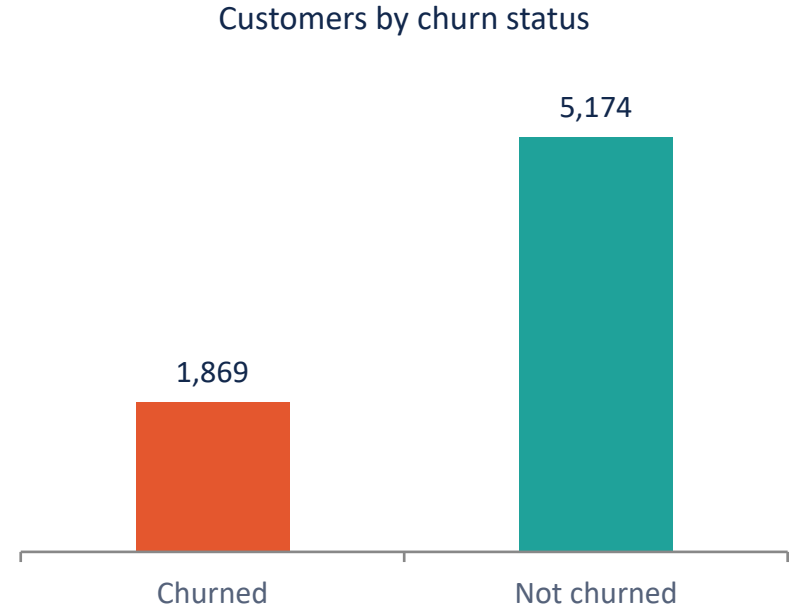
Duplicate rows: 0

```
1 # 5. Check missing values
2
3 missing = df.isnull().sum()
4 missing
```

Exploratory Data Analysis (EDA)

Libraries: Pandas, NumPy, Matplotlib, Seaborn

- 1,869 of 7,043 customers churned (26.54%)
- 5,174 customers did not churn (73.46%)
- Churn by contract: month-to-month 42.71%, one-year 11.27%, two-year 2.83%
- Also compared: gender, senior status, internet service, payment method, partner/dependents, services
- Compared tenure, monthly and total charges by churn; correlation of numeric measures



Data Visualization

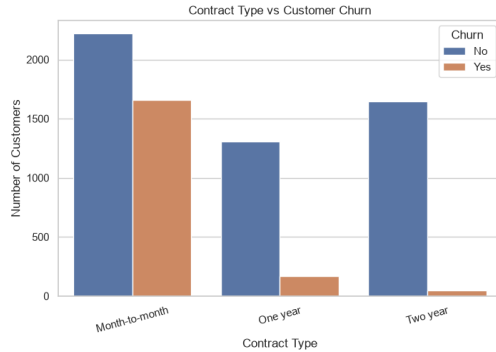


Chart → Contract Type vs Churn

Purpose → Compare churn across contracts

Shows → Churn rate falls with longer contracts

Insight → Month-to-month: 42.71%; two-year: 2.83%

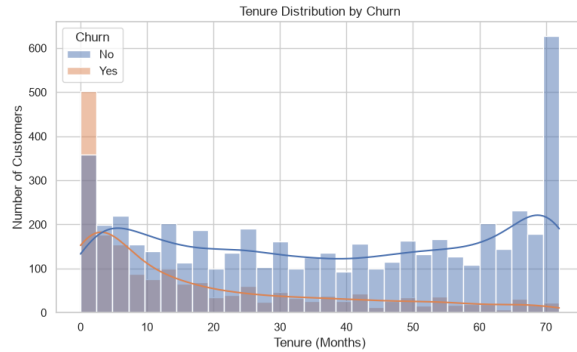


Chart → Tenure Distribution by Churn

Purpose → See how tenure differs by status

Shows → [what chart shows]

Insight → [insight from your chart]

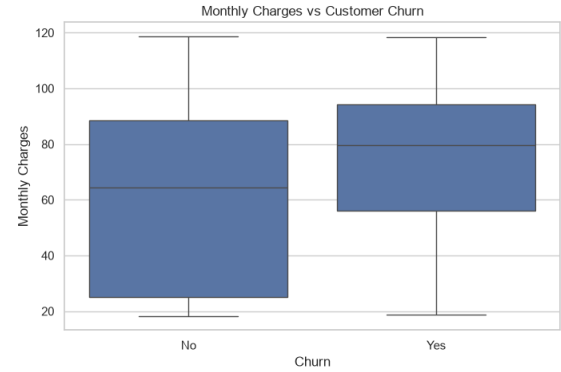


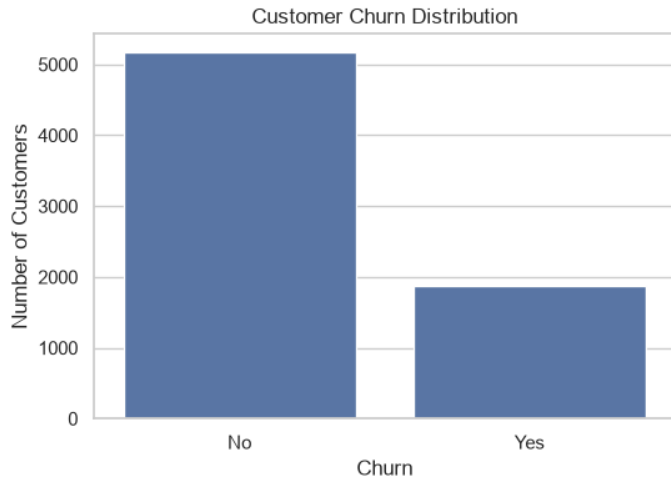
Chart → Monthly Charges vs Churn

Purpose → Compare charges by churn status

Shows → [what chart shows]

Insight → [insight from your chart]

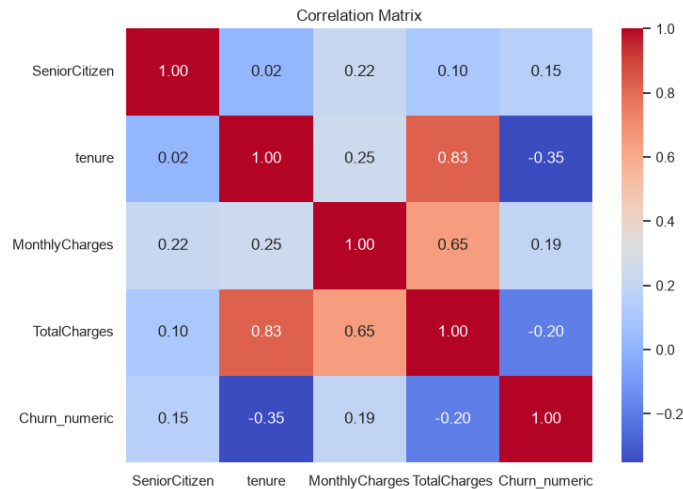
Visualizations: Distribution & Correlation



Customer Churn Distribution

Purpose: show churn count and proportion

Insight: 26.54% churned, 73.46% did not



Correlation Matrix

Purpose: show relationships among numeric measures and churn

Insight: [your observation]

Key Insights, Recommendations & Conclusion

Key Insights

- 26.54% of customers churned (1,869 of 7,043)
- Churn: month-to-month 42.71% vs one-year 11.27% vs two-year 2.83%
- Descriptive results; not proof of causation

Recommendations

- Review retention needs for month-to-month customers (highest observed churn)
- Evaluate any retention action against later churn outcomes

Conclusion

Cleaned and analyzed 7,043 telecom customers, identifying segments where churn is concentrated using Python.

Future Scope

Build predictive churn models and measure the effect of retention interventions.

